

PhytoGraph — cycles 24-26

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PhytoGraph — cycles 24-26

Abstract

Cycles 24-26 executed three focused reopen branches after the Wave 5 final synthesis had already closed the PhytoGraph campaign conservatively. Each branch tested whether a specific missing evidence predicate had become true for one track:

- Cycle 24 tested Track 4, the Domestication Hypergraph, for observed crop/crop-wild-relative BIOCLIM vectors and validation-ready expert comparator rows.
- Cycle 25 tested Track 5, the Chemodiversity Predictor, for accepted-key non-Duke dated taxon-compound evidence.
- Cycle 26 tested Track 6, the Botanical Foundation Model Probe, for a runnable free/open/local model runtime plus local model weights and audited benchmark responses.

All three branches ended with the same determination: `no_new_qualifying_evidence`. No predictor was rerun, no model was executed, and no master prediction or speculation row was promoted. The master `prediction_ledger.tsv` and `speculation_ledger.tsv` remained header-only.

The supplied audit report, labeled cycle 27, validated the post-reopen closure addendum that reconciled these three branches with the earlier Track 1 reopen result. That closure context confirms that none of the four tested reopen predicates for Tracks 1, 4, 5, and 6 became true.

Introduction

PhytoGraph is a six-track research campaign built around a typed, auditable hypergraph of plant biology. Its goal is not to collect known facts into a catalog, but to produce falsifiable predictions and then validate, falsify, or close them cleanly.

By cycles 24-26, the campaign was already in post-closure mode. The Wave 5 final synthesis had recorded conservative outcomes across all six tracks, and the reopen evidence gate from cycle 22 had defined the exact evidence predicates required to reconsider several closed

branches. In this context, “reopen” means a branch may change status only if the missing evidence class is actually present in the workspace and passes the relevant join, validation, and claim-boundary checks.

Three terms recur in this report:

- **Accepted key:** the canonical taxon identifier used by the frozen PhytoGraph substrate, based primarily on the World Flora Online naming layer.
- **CWR:** crop wild relative, a wild taxon related to a crop and potentially relevant to crop improvement.
- **BIOCLIM vector:** a numerical climate summary derived from occurrence or range data and climate rasters such as WorldClim or CHELSA.

Approach

The report was assembled from local workspace artifacts, the supplied audit report, and the append-only `promise_ledger.jsonl`. Direct session-fetch/search tools for the listed session IDs were not available in this environment, so the raw provider-native transcripts for the cycle sessions could not be fetched. This is a record gap. The available trace consists of the cycle reports, data tables, tests, figures, ledger events, and the supplied auditor validation.

The cycle session IDs supplied for traceability were:

Cycle	Researcher	Worker	Auditor
24	ce3702eb-ad17-4e3d-89a6-cabef9b18ded	24546f88-788b-4002-8e7a-8da5e75b2498	a74fbcc7-24a2-4222-bd42-6aef55c80029
25	a6096ee4-f44a-4b22-813b-61d1289fa619	ca7377bb-fe81-4cf5-b409-79e0353610c0	29fae4c5-b7e2-429d-89f5-c099b319a9a3
26	f7ed400c-e196-4dcc-86b8-ff81b7854134	43ad877f-4bc9-4b43-9aba-e5ddf504ebfa	68831f63-3358-429a-8411-f70055c4b161

The local promise-ledger events used for the main chronology were the cycle 24 worker/auditor events 2d03bc31-924d-49f8-9a8f-1fd02f36a1e1 and 8b70833f-f79b-4dc0-b211-e8ae1148e594, the cycle 25 worker/auditor events 9d6e9453-737d-45cf-abea-3f4f5942dd8f and 2d8c84cb-81ac-45b7-9d3e-18f3b8de98f2, and the cycle 26 worker/auditor events b0e8cd52-bdc2-4af8-a85f-efea0da6c6c3 and d4a3ef34-f9fd-4540-a08b-5683508d394c.

Findings

Cycle 24: Track 4 Did Not Recover Climate-Validation Evidence

Cycle 24 tested whether Track 4 could reopen H4 by finding the two evidence classes needed for climate-aware crop substitution validation: observed crop/CWR BIOCLIM vectors and held-out expert comparator rows.

The worker package was `tracks/track4/reports/track4_reopen_bioclim_validation_readiness.md`. It inspected four local artifacts: climate-envelope coverage, crop-wild-relative pairs, a held-out expert crop-source set, and prior Track 4 candidate rows. The result was `no_new_qualifying_evidence`.

The quantitative outcome was:

Evidence class	Result
Climate-envelope staging rows	375
Accepted-key climate joins	36
Crop/CWR pair rows	69
Fully joined crop/CWR rows	3
Held-out expert crop rows	22
Accepted-key held-out crop joins	2
Numeric BIOCLIM vector rows	0
Validation-allowed held-out candidate-level pairs	0

The branch produced `tracks/track4/data/crop_cwr_bioclim_vectors.tsv`, which contains 10 accepted-key crop/CWR climate-coverage attempts. Every row has `bioclim_variable=none_available`, an empty value field, and `aggregation_method=not_computed`. These rows document coverage attempts, not usable climate vectors.

The branch also produced `tracks/track4/data/crop_cwr_validation_pairs.tsv`, which contains 22 held-out crop-source rows. All rows avoid training overlap, but all have `validation_allowed=false` because the available held-out sources are crop-level references rather than candidate-level expert CWR comparator rows.

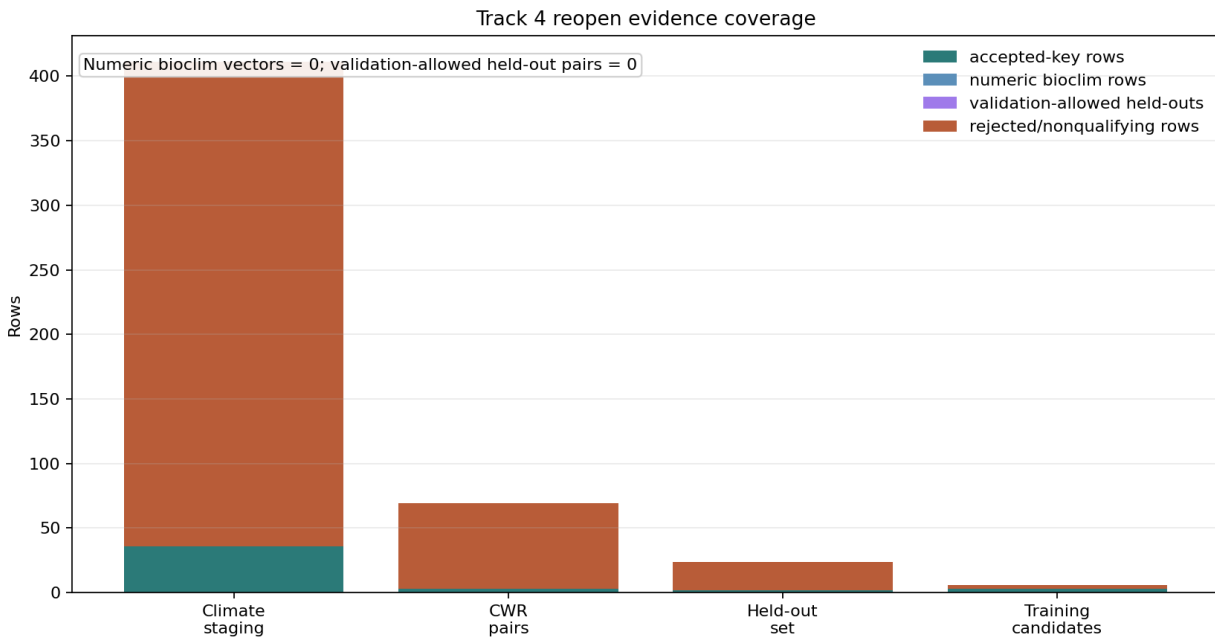


Figure 1: Accepted-key crop/CWR coverage for observed BIOCLIM vectors and held-out expert-comparison rows.

The auditor event `8b70833f-f79b-4dc0-b211-e8ae1148e594` validated this as a non-reopen result. The Crop Substitution Engine was not rerun, and no climate-substitution recommendation, prediction-ledger row, speculation-ledger row, crop-suitability claim, or biological claim was promoted.

Cycle 25: Track 5 Did Not Recover Non-Duke Temporal Chemistry Evidence

Cycle 25 tested whether Track 5 could reopen H5 by finding source-independent, temporally resolved phytochemical evidence. The specific predicate was accepted-key, non-Duke taxon-compound evidence with usable discovery or isolation dates.

The worker package was `tracks/track5/reports/track5_reopen_temporal_chemistry_evidence.md`. It inspected the frozen local chemistry substrate and source diagnostics. The result was again `no_new_qualifying_evidence`.

The key source diagnostic was:

Source class	Candidate rows	Accepted-key rows	Dated rows	Non-Duke rows
Dr. Duke phytochemical rows	103,663	13,867	0	0
ChEBI	0	0	0	0
KNAPSAcK	0	0	0	0
NPASS	0	0	0	0
Native American Ethnobotany Database mirror	0	0	0	0
PROSEA	0	0	0	0
PROTA	0	0	0	0

The canonical holdout matrix contained eight taxa: *Taxus brevifolia*, *Catharanthus roseus*, *Cinchona officinalis*, *Artemisia annua*, *Digitalis purpurea*, *Papaver somniferum*, *Atropa belladonna*, and *Salix alba*. None had validation-allowed non-Duke pre-target or post-target evidence. Only *Artemisia annua* had an accepted key in the frozen substrate, and it still lacked qualifying non-Duke dated taxon-compound rows.

The branch also carried forward the earlier source-dominance result: `source_ablation_results.tsv` had 1,405 baseline prediction rows, but 0 rows under no-Duke, source-density-matched, and screening-count-matched variants. The cycle 25 branch added no evidence that changed that condition.

The auditor event `2d8c84cb-81ac-45b7-9d3e-18f3b8de98f2` validated the package. The chemodiversity predictor was not rerun. No phytochemical novelty, clinical efficacy, preparation, dosage, safety, bioactivity, master prediction, or speculation claim was promoted.

Cycle 26: Track 6 Did Not Find A Runnable Local Model

Cycle 26 tested whether Track 6 could reopen H6 under the free/open/local constraint. The required predicate was at least one local model runtime plus compatible local model weights, producing audited deterministic responses for the static botanical probe.

The worker package was `tracks/track6/reports/track6_reopen_local_model_execution.md`. It inspected local Python packages, local binaries, and workspace model-weight files. The result was `no_new_qualifying_evidence`.

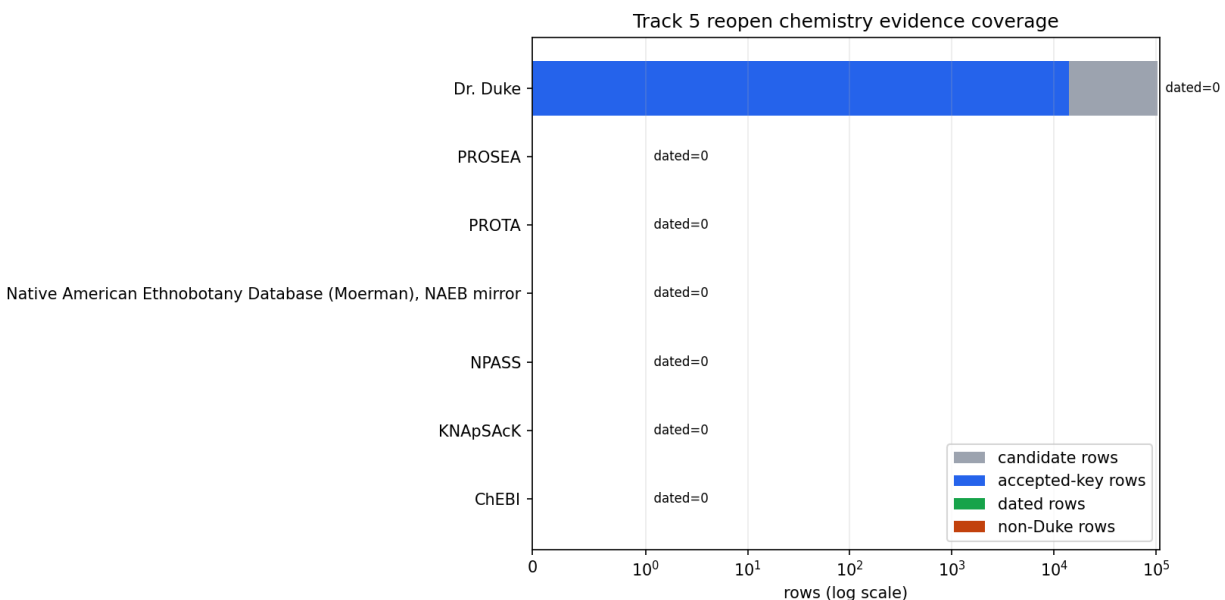


Figure 2: Accepted-key and dated taxon-compound coverage by source for the Track 5 reopen assessment.

The availability table contained 12 rows. Every inspected runtime or asset was unavailable or non-runnable:

Runtime or asset class	Result
Python packages <code>llama_cpp</code> , <code>transformers</code> , <code>torch</code> , <code>ctransformers</code> , <code>gpt4all</code> , <code>onnxruntime</code> , <code>sentencepiece</code>	Missing
Binaries <code>ollama</code> , <code>llama-cli</code> , <code>llamafire</code> , <code>gpt4all</code>	Missing from PATH
Workspace model files matching <code>*.gguf</code> , <code>*.safetensors</code> , <code>pytorch_model*.bin</code> , or <code>model*.onnx</code>	Missing

The static benchmark remained intact: 210 question rows across seven categories. The reopen package wrote explicit `not_run_no_local_model` rows instead of fabricating responses.

Probe category	Static questions	Runnable responses	Scored responses
convergence_	30	0	0
detection	30	0	0
ghost_partner_	30	0	0
reasoning	30	0	0
hybrid_pedigree	30	0	0
phytochemistry_	30	0	0
safety	30	0	0
region_	30	0	0
conditional	30	0	0
synonym_	30	0	0
confusion	30	0	0

Probe category	Static questions	Runnable responses	Scored responses
toxicity_ lookalike_media_ scope	30	0	0

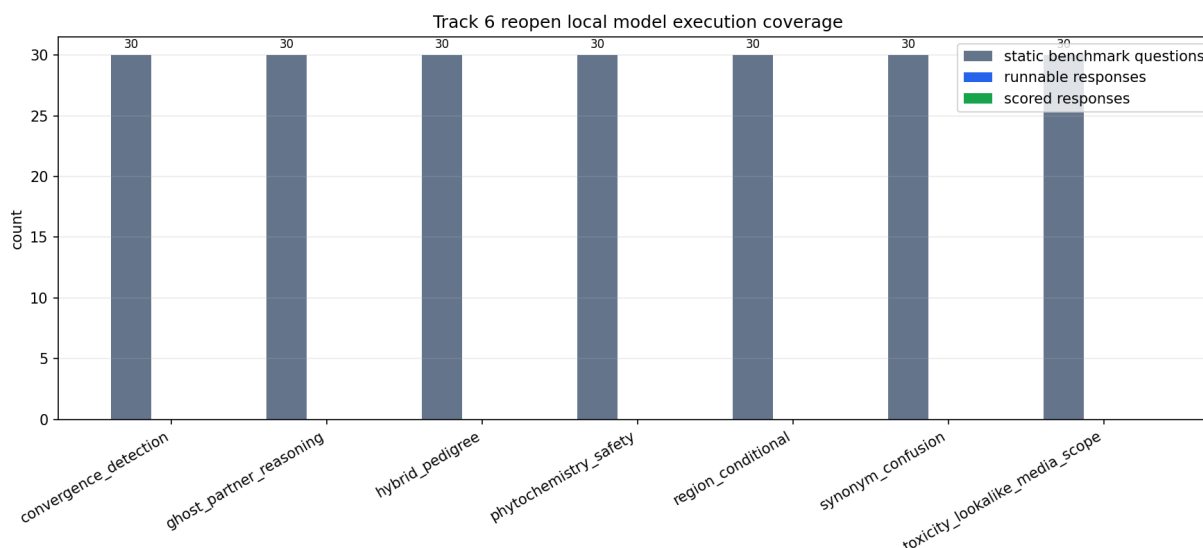


Figure 3: Track 6 reopen coverage by probe category, separating static benchmark size from executed and scored responses.

The auditor event d4a3ef34-f9fd-4540-a08b-5683508d394c validated this as a non-reopen result. No remote, paid, key-gated, or hosted provider execution was configured. No model-performance, leaderboard, vendor-comparison, toxicity-safety, failure-rate, prediction-ledger, or speculation-ledger claim was promoted.

Supplied Closure Context: The Four Reopen Branches Remained Below Promotion Threshold

The supplied audit report validates a post-reopen closure package labeled cycle 27. Because the requested range is cycles 24-26, this closure package is context rather than part of the main chronology. It matters because it reconciles the cycle 24-26 branches with the earlier Track 1 reopen branch from cycle 23.

The closure addendum is reports/reopen/reopen_closure_addendum.md, and the machine-readable status table is data/reopen/reopen_closure_status.tsv.

The consolidated status table is:

Track	Reopen result	Quantitative blocker	Master-ledger action
Track 1	evidence_added_but_threshold_not_met	6 accepted-key event-shaped rows across 3 accepted-key taxa, dominated by one source; no additional 30 accepted-key rows for source-density or family-size controls	No master prediction or speculation row
Track 4	no_new_qualifying_evidence	0 observed numeric BIOCLIM vectors and 0 validation-allowed held-out candidate-level crop/CWR comparison pairs	No master prediction or speculation row
Track 5	no_new_qualifying_evidence	0 accepted-key non-Duke taxon-compound rows with usable discovery_or_isolation_year; no-Duke/source-matched variants remain at 0 prediction rows	No master prediction or speculation row
Track 6	no_new_qualifying_evidence	0 runnable runtime-weight pairings, 0 executed model responses, and 0 scored responses	No master prediction or speculation row

The supplied auditor event 83116d67-3a6c-43d1-aa5f-c06743fcd2ef validated the closure package. The audit reported 13 focused tests passing, promise_check exiting 0 with inherited warnings only, and org_check exiting 0 with inherited root-layout warnings only. The audit also recorded promise_ledger.jsonl at 205 lines after the auditor append.

Discussion

Cycles 24-26 did not change the scientific status of the campaign. They made the closure record more precise.

Track 4 moved from a general data-limited condition to a specific blocker: climate-aware validation cannot run because there are zero observed numeric BIOCLIM vectors and zero validation-allowed candidate-level expert comparator pairs. The branch documents accepted-key coverage attempts, but those attempts do not become climate evidence.

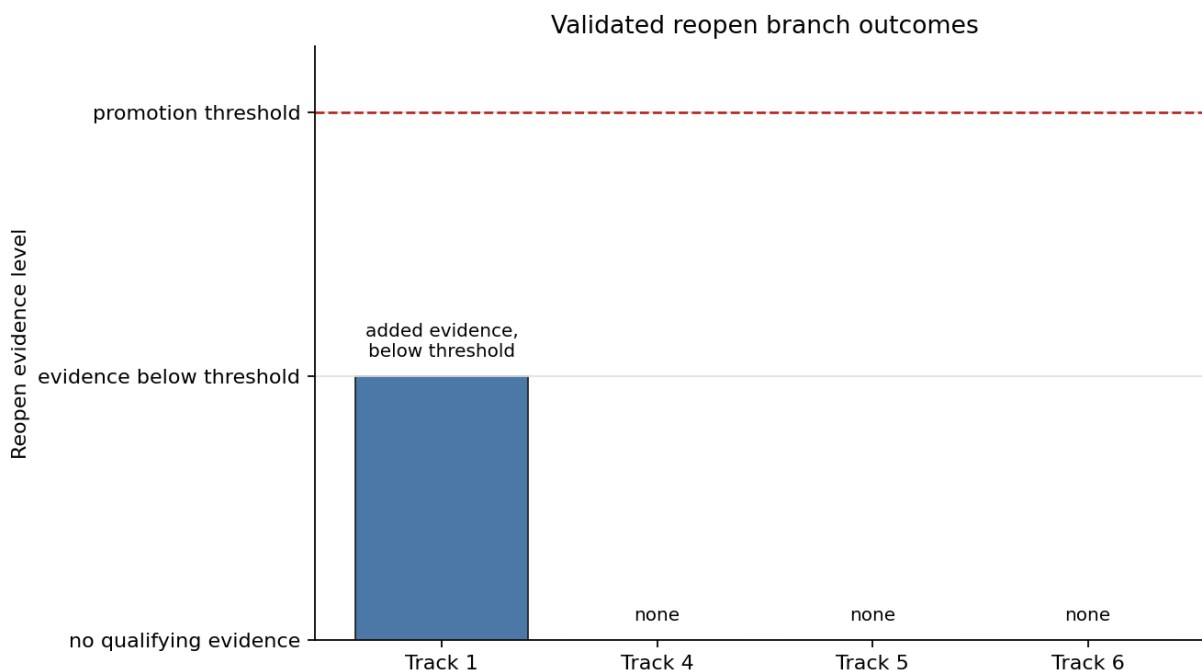


Figure 4: Validated reopen branch outcomes by track.

Track 5 confirmed that the source-bias problem remained unresolved. The local chemistry substrate contains many Duke-derived rows, including accepted-key retained rows, but it contains zero accepted-key non-Duke dated taxon-compound rows. Since the target validation protocol depends on temporally frozen, source-independent recovery of known phytochemical discoveries, the predictor could not be rerun.

Track 6 confirmed that the free/open/local execution constraint remained binding. The static benchmark and deterministic scoring infrastructure exist, but without a runnable local model and local weights, model error rates are undefined. The branch correctly recorded skipped rows rather than treating the benchmark itself as a model evaluation.

The common decision across all three cycles was non-promotion. This means the track-local diagnostic tables remain useful as audit artifacts, but the master ledgers remain reserved for claims satisfying the campaign’s validation and ablation contracts.

Open Questions

The future reopen predicates are now explicit.

Track 4 can reopen only if occurrence-backed crop and CWR coordinates yield accepted-key BIOCLIM summaries, and disjoint candidate-level expert comparator rows exist before climate-substitution scoring.

Track 5 can reopen only if non-Duke taxon-compound detections have accepted taxon joins, compound identifiers, usable dates, family spread, and controls that preserve signal without Duke source dominance.

Track 6 can reopen only if approved local model weights and a free/open/local runtime produce audited deterministic responses with scorer diagnostics and nonzero scored-response coverage.

Track 1 remains outside the main cycle 24-26 chronology, but the supplied closure context records its future predicate: independent local/open event-shaped reticulation evidence must join to accepted keys, clear canonical recovery thresholds, and provide enough additional accepted-key rows for source-density and family-size controls.

References

- [4] The World Flora Online Consortium et al., “World Flora Online Plant List December 2025,” Zenodo, 2025. <https://doi.org/10.5281/zenodo.18007552> (accessed 2026-05-17).
- [38] James A. Duke, “Dr. Duke’s Phytochemical and Ethnobotanical Databases,” Ag Data Commons, 2023. https://agdatacommons.nal.usda.gov/articles/dataset/Dr_Duke_s_Phytochemical_and_Ethnobotanical_Databases/24660351 (accessed 2026-05-17).
- [40] KNApSAcK Family, “KNApSAcK Core System,” KNApSAcK Family Databases, 2026. <http://www.knapsackfamily.com/KNApSAcK/> (accessed 2026-05-17).
- [41] Zeng et al., “NPASS: Natural Product Activity and Species Source Database,” NPASS, 2026. <https://bidd.group/NPASS/> (accessed 2026-05-17).
- [42] EMBL-EBI, “ChEBI: Chemical Entities of Biological Interest,” EMBL-EBI, 2026. <https://www.ebi.ac.uk/chebi/> (accessed 2026-05-17).
- [43] PROTA Foundation, “Plant Resources of Tropical Africa,” PROTA4U, 2026. <https://prota.prota4u.org/> (accessed 2026-05-17).
- [44] PlantUse, “Plant Resources of South-East Asia (PROSEA),” PlantUse, 2026. <https://uses.plantnet-project.org/en/PROSEA> (accessed 2026-05-17).

Appendix: Implementation Details

Source Inventory

Source	Date or cycle	Contents	Timeline role
Cycle 24 worker event 2d03bc31-924d-49f8-9a8f-1fd02f36a1e1	2026-05-18, cycle 24	Track 4 bioclim validation-readiness package	Produced cycle 24 non-reopen evidence package
Cycle 24 auditor event 8b70833f-f79b-4dc0-b211-e8ae1148e594	2026-05-18, cycle 24	Validated Track 4 package	Superseded worker event and confirmed no_new_qualifying_evidence
Cycle 25 worker event 9d6e9453-737d-45cf-abea-3f4f5942dd8f	2026-05-18, cycle 25	Track 5 non-Duke temporal chemistry package	Produced cycle 25 non-reopen evidence package
Cycle 25 auditor event 2d8c84cb-81ac-45b7-9d3e-18f3b8de98f2	2026-05-18, cycle 25	Validated Track 5 package	Superseded worker event and confirmed no_new_qualifying_evidence
Cycle 26 worker event b0e8cd52-bdc2-4af8-a85f-efea0da6c6c3	2026-05-18, cycle 26	Track 6 local model execution package	Produced cycle 26 non-reopen evidence package

Source	Date or cycle	Contents	Timeline role
Cycle 26 auditor event d4a3ef34-f9fd-4540-a08b-5683508d394c	2026-05-18, cycle 26	Validated Track 6 package	Superseded worker event and confirmed no_new_qualifying_evidence
Supplied audit event 83116d67-3a6c-43d1-aa5f-c06743fcd2ef	2026-05-18, cycle 27	Post-reopen closure validation	Context for reconciling Tracks 1, 4, 5, and 6

Code Organization

Area	Files	Role
Track 4 reopen	tracks/track4/scripts/build_reopen_bioclim_validation.py, tracks/track4/scripts/plot_reopen_bioclim_coverage.py, tests/test_track4_reopen_bioclim_validation.py	Build and verify bioclim validation-readiness artifacts
Track 5 reopen	tracks/track5/scripts/build_reopen_temporal_chemistry_evidence.py, tracks/track5/scripts/plot_reopen_temporal_chemistry_coverage.py, tests/test_track5_reopen_temporal_chemistry.py	Build and verify non-Duke temporal chemistry evidence artifacts
Track 6 reopen	tracks/track6/scripts/build_reopen_local_model_execution.py, tracks/track6/scripts/plot_reopen_execution_coverage.py, tests/test_track6_reopen_local_model_execution.py	Build and verify local model availability, not-run response rows, and scoring diagnostics
Closure context	reports/reopen/scripts/plot_reopen_branch_outcomes.py, tests/test_reopen_closure_addendum.py	Build and verify the post-reopen closure outcome figure and table

File Counts

Path	Rows or lines
tracks/track4/reports/track4_reopen_bioclim_validation_readiness.md	55 lines
tracks/track4/data/crop_cwr_bioclim_vectors.tsv	11 lines, 10 data rows
tracks/track4/data/crop_cwr_validation_pairs.tsv	23 lines, 22 data rows

Path	Rows or lines
tracks/track4/data/track4_reopen_join_diagnostics.tsv	5 lines, 4 data rows
tracks/track4/scripts/build_reopen_bioclim_validation.py	255 lines
tracks/track4/scripts/plot_reopen_bioclim_coverage.py	67 lines
tests/test_track4_reopen_bioclim_validation.py	125 lines
tracks/track5/reports/track5_reopen_temporal_chemistry_evidence.md	54 lines
tracks/track5/data/non_duke_temporal_taxon_compound_evidence.tsv	1 line, 0 data rows
tracks/track5/data/track5_reopen_temporal_holdout_matrix.tsv	9 lines, 8 data rows
tracks/track5/data/track5_reopen_source_diagnostics.tsv	8 lines, 7 data rows
tracks/track5/scripts/build_reopen_temporal_chemistry_evidence.py	268 lines
tracks/track5/scripts/plot_reopen_temporal_chemistry_coverage.py	51 lines
tests/test_track5_reopen_temporal_chemistry.py	135 lines
tracks/track6/reports/track6_reopen_local_model_execution.md	58 lines
tracks/track6/data/local_model_availability_reopen.tsv	13 lines, 12 data rows
tracks/track6/data/local_model_probe_responses.tsv	211 lines, 210 data rows
tracks/track6/data/local_model_probe_scoring_diagnostics.tsv	8 lines, 7 data rows
tracks/track6/scripts/build_reopen_local_model_execution.py	304 lines
tracks/track6/scripts/plot_reopen_execution_coverage.py	49 lines
tests/test_track6_reopen_local_model_execution.py	81 lines
reports/reopen/reopen_closure_addendum.md	81 lines
data/reopen/reopen_closure_status.tsv	5 lines, 4 data rows
prediction_ledger.tsv	1 line, header only
speculation_ledger.tsv	1 line, header only
promise_ledger.jsonl	205 lines at scan time

Test Results

The supplied audit reported:

```
python3 -m pytest -q tests/test_reopen_closure_addendum.py tests/test_wave4_postmerge_integration.py
↳ tests/test_barrier4_closure_integration.py
13 passed in 0.67s
```

It also reported:

```
python3 -m long_exposure.tools.promise_check /home/jeremyad55/workspace
exit 0, 205 events, inherited warnings only
```

```
python3 -m long_exposure.tools.org_check /home/jeremyad55/workspace
exit 0, inherited root-layout warnings only
```

The branch-level auditor events additionally report focused pytest validation for the Track 4, Track 5, and Track 6 reopen packages.

Cross-Reference Map

From	To	Meaning
data/reopen/reopen_branch_matrix.tsv	Cycle 24 Track 4 package	Defines the H4 predicate tested in cycle 24
tracks/track4/data/crop_cwr_bioclim_vectors.tsv	tracks/track4/reports/track4_reopen_bioclim_validation_readiness.md	Provides the zero-numeric-BIOCLIM basis for non-reopen
tracks/track4/data/crop_cwr_validation_pairs.tsv	tracks/track4/reports/track4_reopen_bioclim_validation_readiness.md	Provides the zero-validation-allowed-comparator basis for non-reopen
data/reopen/reopen_branch_matrix.tsv	Cycle 25 Track 5 package	Defines the H5 predicate tested in cycle 25
tracks/track5/data/non_duke_temporal_taxon_compound_evidence.tsv	tracks/track5/reports/track5_reopen_temporal_chemistry_evidence.md	Provides the zero-row non-Duke temporal evidence basis for non-reopen
tracks/track5/data/track5_reopen_temporal_holdout_matrix.tsv	tracks/track5/reports/track5_reopen_temporal_chemistry_evidence.md	Documents eight canonical holdouts, all validation-disallowed
data/reopen/reopen_branch_matrix.tsv	Cycle 26 Track 6 package	Defines the H6 predicate tested in cycle 26
tracks/track6/data/local_model_availability_reopen.tsv	tracks/track6/reports/track6_reopen_local_model_execution.md	Documents zero runnable runtime-weight pairings
tracks/track6/data/local_model_probe_responses.tsv	tracks/track6/reports/track6_reopen_local_model_execution.md	Preserves 210 not-run benchmark rows without fabricating responses
Cycles 23-26 branch packages	reports/reopen/reopen_closure_addendum.md	Reconciles Tracks 1, 4, 5, and 6 into validated non-promotion
prediction_ledger.tsv and speculation_ledger.tsv	All reopen reports	Remain header-only because no predicate became true

Manifest Update

MANIFEST.md was updated for report_cycles_24-26. The existing ## Key Files section was preserved verbatim. The mutable sections now list the cycle 24-26 scripts, data artifacts, cumulative counts, and cross-references.

Record Gaps

The named cycle session IDs were supplied, but no session-search or session-fetch tool was available in this environment. The report therefore relies on local workspace artifacts, promise-ledger events, tests, figures, and the supplied audit report. This gap does not change the reported branch outcomes because each branch is represented by local artifacts and auditor-validated ledger events.

Coherence Review

The report was reviewed as a self-contained narrative. The main correction during review was to keep the supplied cycle-27 closure audit separate from the cycle 24-26 chronology while still using it as validated context for the final non-promotion decision.